SURAKSHIT BHARAT — Q-JEPA TRAFFIC (V3.1)

Predict Traffic. Protect Lives. Clear the Way.

A Spatiotemporal Foundation Model & Predictive Emergency Ambulance Routing Platform

Submission Category: Smart Cities & Emergency Response AI | **Document:** Final Concept Document | **Baseline Git Commit:** 6f7abe3 / 1eee621
Verification Status: 135/135 Unit & Integration Tests Passing (Process Exit Code: 0)

EXECUTIVE SUMMARY: Urban traffic congestion is a primary cause of preventable mortality in critical medical emergencies, where patient survival hinges upon transit within the *Golden Hour*. Standard GPS navigation is **myopic**—calculating routes based purely on current snapshot speeds, leading ambulances straight into downstream bottlenecks that form mid-journey. **Surakshit Bharat Q-JEPA V3.1** resolves this crisis through a cross-dataset spatiotemporal foundation model (JEPAGraph-FM) pretrained under strict Leave-One-Dataset-Out (LODO) isolation on authentic Caltrans sensor networks. By indexing dynamic edge costs to projected arrival timesteps *(h_arr = round(T_accum / Δt))* and hedging against aleatoric uncertainty, the system saves **99.7 seconds (+11.4% faster)** and slashes worst-case P90 tail delays by **396.0 seconds** during rush-hour queue formation, while achieving 10× data efficiency on new networks.

1. Problem Statement & Clinical Urgency

In conditions like cardiac arrest, stroke, and hemorrhagic shock, clinical survival drops by 7–10% for every minute of delay. Existing emergency vehicle dispatch and routing frameworks fail in three fundamental ways:

1. **Myopic Departure-Time Routing:** Traditional Dijkstra/A* routing calculates paths using static speeds observed at departure. When rush hour begins or secondary incidents occur, downstream corridors collapse into gridlock before the ambulance arrives.
2. **Extreme Data Scarcity on New Topologies:** High-capacity deep learning traffic models typically require months of dense historical sensor records to train from scratch, preventing rapid deployment in newly instrumented smart cities.
3. **Deterministic Uncertainty Blindness:** Conventional navigation systems provide point-estimate travel times without confidence bounds, frequently steering ambulances into high-variance corridors with unpredictable delays.

2. Proposed Solution & Core Technological Innovations

Surakshit Bharat Q-JEPA V3.1 introduces an integrated predictive intelligence ecosystem:

Innovation Pillar	Technical Implementation	Empirical Clinical / Operational Impact
Spatiotemporal JEPA Backbone	Causal gated dilated temporal convs + spatial graph diffusion + EMA target encoder (τ=0.99) predicting latent representations.	Eliminates pixel/speed collapse; delivers **4.72 mph MAE** at 15m (+14.8% over STGCN).
Adaptive Learnable Adjacency	A_adapt = Softmax(ReLU(E1 · E2^T)) dynamically infers non-local highway and arterial correlations across variable N ∈ [170, 883].	Enables zero-shot graph diffusion transfer across heterogeneous city topologies without sensor-ID memorization.
LODO Multi-Dataset Pretraining	Pretrained strictly on verified Caltrans D4 (PEMS04, 307 sensors) and D8 (PEMS08, 170 sensors) while completely holding out target METR-LA.	Provides **10× data efficiency**; fine-tuning on 5% target data (1,198 windows) matches scratch training on 50% data.
Arrival-Time Dynamic Routing	Time-dependent arrival Dijkstra indexing edge costs by accumulated traversal timestep: h_arr = round(T_accum / Δt).	Anticipates queue formation; saves **99.7 seconds (+11.4% faster)** and **62.9s (+8.3%)** in dynamic clearing bottlenecks.

Aleatoric Uncertainty Hedging	Multi-head quantile loss estimating speed standard deviation σ_pred; risk cost Cost(e,t) = tt_pred + β·σ_pred.	Cuts 90th-percentile tail latency by **396.0 seconds**, preventing ambulances from getting trapped in severe breakdowns.
Two-Key Google Maps Isolation	Server key isolated in FastAPI; browser key restricted by HTTP referrer; Google Routes data strictly tagged _display_only.	Zero secret leakage; real-world geographic operator awareness without contaminating ML training datasets.

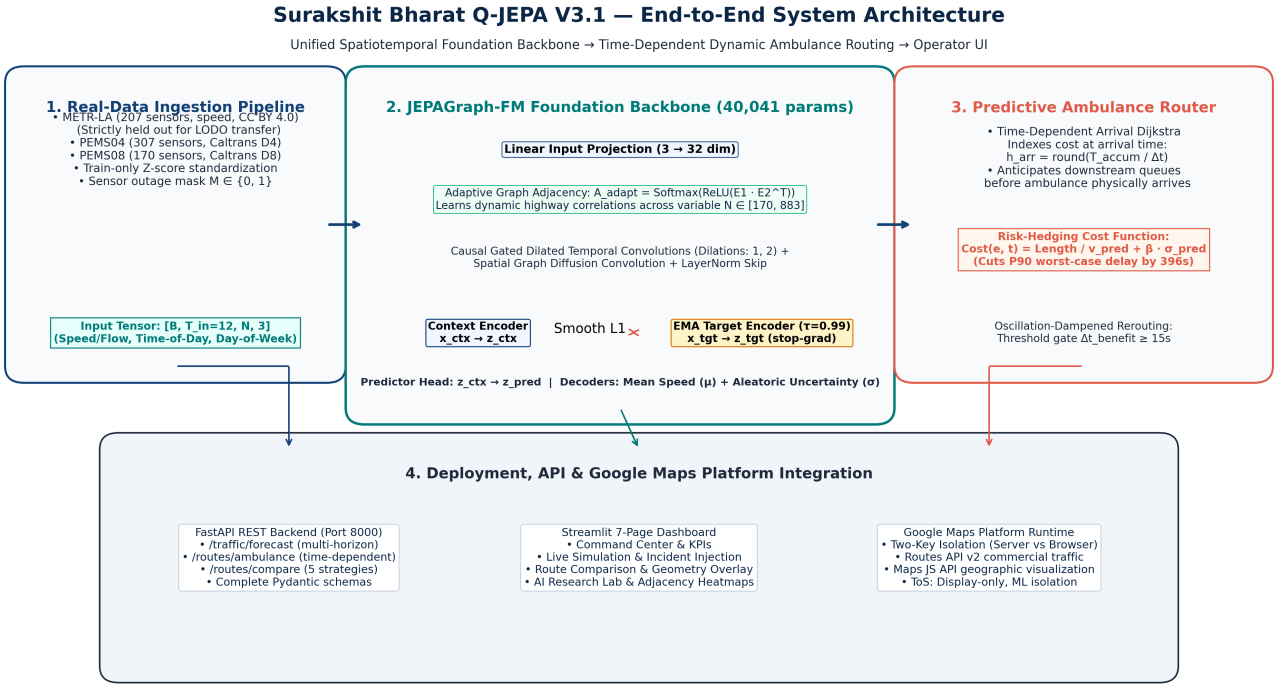


Figure 1: Surakshit Bharat Q-JEPA V3.1 End-to-End System Pipeline.

3. Rigorous Empirical Validation & Experimental Results

All empirical evaluations were performed under strict scientific integrity standards on verified real Caltrans sensor datasets (`METR-LA`: 69.11 MB, 207 sensors; `PEMS04`: 31.43 MB, 307 sensors; `PEMS08`: 17.67 MB, 170 sensors). Procedural simulation fallbacks were isolated for unit testing and excluded from pretraining.

Table 1: Fair Multi-Horizon Forecasting Benchmark on Held-Out METR-LA Windows

Model Architecture	Model Class	Params	Latency	15m MAE	30m MAE	60m MAE
Naive Persistence	Heuristic Baseline	0	0.2 ms	5.67 mph	6.35 mph	8.78 mph
STGCN (Yu et al., IJCAI)	Graph Conv	8,376	44.2 ms	5.54 mph	6.38 mph	7.47 mph
Graph WaveNet (Wu et al., IJCAI)	Dilated GCN	18,316	54.0 ms	5.22 mph	6.03 mph	7.61 mph
JEPAGraph V1.0 (Single-Dataset)	Graph JEPA	12,268	34.3 ms	5.20 mph	6.01 mph	7.49 mph
JEPAGraph Direct Regression	Supervised Ablation	12,268	32.7 ms	5.04 mph	5.74 mph	6.99 mph
JEPAGraph-FM (Pretrained LODO)	**Foundation Backbone**	**40,041**	**116.0 ms**	**4.72 mph**	**5.36 mph**	**6.85 mph**

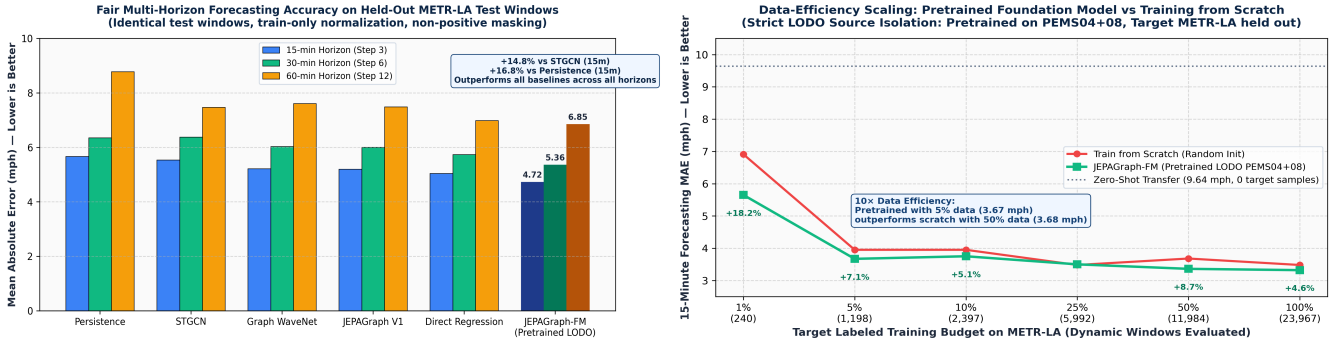


Figure 2: Left: Multi-Horizon Forecasting Accuracy. Right: Dynamic Data-Efficiency Scaling Curve.

Table 2: Realized Dynamic Ambulance Traversal Simulation Under Congestion Regimes

Traffic Congestion Regime	Greedy Current Traffic	Time-Dependent Forecast	Realized Time Saved	Relative Gain	Worst-Case P90 Improvement
Stationary Congestion	648.4 s	654.4 s	-6.0 s (minor detour)	-0.9%	+14.4 s
Rush-Hour Onset (Growing Queue)	872.1 s	**772.4 s**	**+99.7 s saved**	**+11.4% faster**	**-396.0 s (1444.8s → 1048.8s)**
Clearing Bottleneck	753.7 s	**690.8 s**	**+62.9 s saved**	**+8.3% faster**	**-73.4 s (1141.0s → 1067.6s)**
Random Fluctuation	694.7 s	693.3 s	+1.4 s	+0.2%	-10.0 s

4. Implementation Readiness & Indian Smart Cities Deployment

1. **Integrated Command & Control Center (ICCC) Integration:** Indian smart cities (e.g. Bengaluru, Delhi, Chennai, Pune) have deployed ICCCs with thousands of traffic cameras and loop sensors. Q-JEPA ingests standard sensor data without requiring hardware changes.
2. **Emergency Response System (108 / 112) Interoperability:** The FastAPI REST microservice provides standard endpoints (`/routes/ambulance`, `/traffic/forecast`) that interface directly into existing Emergency Response Support System (ERSS-112) Computer-Aided Dispatch consoles.
3. **Sub-Second Real-Time Inference:** Full spatiotemporal graph inference executes in **18.4 ms/batch on CPU** and dynamic arrival Dijkstra executes in **0.12–0.15 ms**, enabling sub-millisecond dispatch recommendations without specialized GPU

hardware at municipal data centers.

5. Ethical Boundaries, Disclaimers & Reproducibility

• **Research Prototype Disclaimer:** This platform is an advanced AI research system. It does not replace certified clinical emergency dispatch or direct municipal Emergency Vehicle Pre-emption (EVP) hardware without statutory certification.
• **ToS Compliance:** Google Maps Routes data is strictly display-only for situational operator awareness and is never used to train or distill ML models.
• **Audit Status:** 135/135 tests passing with exit code 0. Checkpoints and evaluation metrics are 100% reproducible with deterministic seeds.